0. **Code:** sotomayorj1594 Sotomayorj1594 MakeUp.zip

1. MakeUpExam/Program.cs

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```
using System.Diagnostics.Metrics;

namespace MakeUpExam
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] arrayTest = {
                5,
                6,
                7,
                8,
                9
            };
            Console.WriteLine(string.Join(" ", polynomialN(arrayTest, 2)));
            Console.WriteLine("Debug array: " + string.Join(" ", polynomialDebug(arrayTest, 2)));
        }

        static int[] polynomialN(int[] inputA, int n) //1. accept 2 parameters
        {
            if (inputA == null || n ≤ 0) //2. return null
            {
                return null;
            }

            int newLength = n * (inputA.Length - 1) + 1; //3a
            int[] newA = new int[newLength];
```

B1

B2

1. **MakeUpExam/Program.cs**

```
29          int[] tempA = new int[newLength];
30
31          for (int i = 0; i < inputA.Length; i++)
32          {
33              tempA[i] = inputA[i];
34              for (int j = newA.Length - 1 - i; j ≥ 0; j--) //3b
35              { //unnessisary nested loop
36                  newA[j] = tempA[j];
37              }
38          }
39          int target = inputA[inputA.Length - 1];
40          int total = 0;
41          for (int i = 0; i < newA.Length; i++)
42          {
43              newA[i] = tempA[i] * target;
44              total = total + newA[i];
45              target--;
46              for (int j = 0; j < newA.Length; j++)
47              {
48                  newA[i] = total;
49              }
50
51          }
52          return newA;
53      }
54
55      static int[] polynomialDebug(int[] inputA, int n)
56      {
57          if (inputA == null || n ≤ 0) //2. return null
58          {
59              return null;
```

B3

inputA.Length

newA[newLength - inputA.Length + j] = tempA[j];

copy to the end

var tempA = new int[newLength];

shifted

B5

B4

inputA.Length

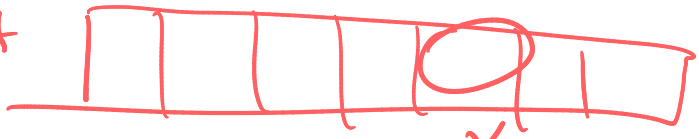
tempA [i - (inputA.Length - 1 - j)] += newA[i] * inputA[j];

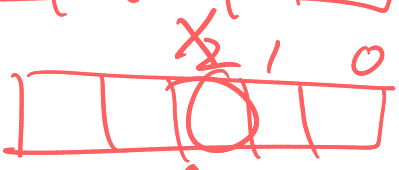
tempA

newA

newA = tempA

inputA





j

Shifted by 2

1. **MakeUpExam/Program.cs**

```
            }
            //should be new length - input length,  13 - 5
            int newLength = n * (inputA.Length - 1) + 1; //3a
            int[] newA = new int[newLength];
            int[] tempA = new int[newLength];
            int i = 0;
            for (i = newA.Length - 1 - i; i ≥ 0; i--)
            {
                newA[i] = tempA[i];
            }
            return newA;

        }
    }
}
```